

Lesson plan (Template –2)

Name of the Programme:	B.Tech(H)	
Title of the Course:	Data Structures	
Course code	CS1006	
Semester:	II	
Names of the Course Instructor(s)	Prof. Kalpana Devi S	
Course credit	No. of hours per week (L + T + P)	Total no. of Teaching hours
04	2+1+2	30L+15T+30P=75 Hours

Session	Module No.	Topic	Pedagogy /Activities	Pre-reading /reference
1	1	Strategies for choosing the appropriate data structure. Abstract data types.	Interactive Lecture	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
2		Stack Memory: Compile Time, Static memory allocation	Interactive Lecture	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
3		Exploration of Recursion Concepts.	Active Learning-Tutorial	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
4		Program to print pyramid pattern	Hands- on Demo	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
5		Program to print Decimal to Binary Conversion.	Hands- on Demo	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.

6		Heap Memory-Runtime memory allocation.	Interactive Lecture	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
7		Heap Memory-Dynamic memory allocation.	Interactive Lecture	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
8		Solving Tower of Hanoi Concept.	Active Learning-Tutorial	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
9		Program to Solve Tower of Hanoi Concept.	Hands- on Demo	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
10		Other Recursion Programs	Hands- on Demo	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
11		Arrays–Revision of basic operations	Interactive Lecture	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
12		Implementation of basic operations – Create, Insert, Traverse, Delete, Search and Update	Interactive Lecture	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
13		Revision on Array Operations	Active Learning-Tutorial	E. Horowitz and S. Sahni, <i>Fundamentals</i>

				<i>of Data Structures in C, 2nd ed.</i>
14		Program to implement Array and its Operations.	Hands- on Demo	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C, 2nd ed.</i>
15		Program to implement Array and its Operations	Hands- on Demo	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C, 2nd ed.</i>
16		Linked lists & its Implementation (SLL)	Interactive Lecture	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C, 2nd ed.</i>
17		Operations on Singly Linked Lists	Interactive Lecture	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C, 2nd ed.</i>
18	2	Discussion on Singly Linked List programming and Applications	Active Learning-Tutorial	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C, 2nd ed.</i>
19		Program to implement Singly Linked List and its operations	Hands- on Demo	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C, 2nd ed.</i>
20		Program to implement Singly Linked List and its operations	Hands- on Demo	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C, 2nd ed.</i>

21		Doubly Linked lists	Interactive Lecture	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
22		Operations on Doubly Linked lists	Interactive Lecture	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
23		Discussion on Doubly Linked List programming and Applications	Active Learning-Tutorial	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
24		Program to implement Doubly Linked List and its operations	Hands- on Demo	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
25		Program to implement Doubly Linked List and its operations	Hands- on Demo	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
26		Circular Singly Linked lists	Interactive Lecture	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
27		Circular Doubly Linked lists	Interactive Lecture	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
28		Discussion on Circular Linked List programming and Applications	Active Learning-Tutorial	E. Horowitz and S. Sahni, <i>Fundamentals</i>

				<i>of Data Structures in C, 2nd ed.</i>
29		Program to implement Circular Singly Linked List and its operations	Hands- on Demo	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C, 2nd ed.</i>
30		Program to implement Circular Doubly Linked List and its operations	Hands- on Demo	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C, 2nd ed.</i>
31		Introduction to Stacks	Interactive Lecture	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C, 2nd ed.</i>
32		Operations on Stacks – Implementation	Interactive Lecture	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C, 2nd ed.</i>
33	3	Stack and its operations using arrays	Active Learning-Tutorial	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C, 2nd ed.</i>
34		Write a Program to Convert Infix to Postfix Conversion using Stack	Hands- on Demo	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C, 2nd ed.</i>
35		Write a Program to Convert Infix to Postfix Conversion using Stack	Hands- on Demo	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C, 2nd ed.</i>

36		Implementation of Stack basic operations using Linked List	Interactive Lecture	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
37		Introduction to Queues	Interactive Lecture	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
38		Queue and its operations using arrays	Active Learning-Tutorial	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
39		Write a program to implement Queue and its operations - Enqueue	Hands- on Demo	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
40		Write a program to implement Queue and its operations – Dequeue and display	Hands- on Demo	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
41		Operations on Queues – Implementation	Interactive Lecture	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
42		Implementation of Queue applications using arrays	Interactive Lecture	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
43		Application of Stacks and Queues	Active Learning-Tutorial	E. Horowitz and S. Sahni, <i>Fundamentals</i>

				<i>of Data Structures in C, 2nd ed.</i>
44		Write a Program to implement Postfix Evaluation using Stack	Hands- on Demo	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C, 2nd ed.</i>
45		Write a program to implement First Come First Serve CPU Scheduling Algorithm	Hands- on Demo	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C, 2nd ed.</i>
46		Linear Search, Binary Search	Interactive Lecture	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C, 2nd ed.</i>
47		Bubble Sort, Selection Sort, Insertion Sort	Interactive Lecture	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C, 2nd ed.</i>
48	4	Discussion on Iterative and Recursive approach on Linear and Binary search	Active Learning-Tutorial	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C, 2nd ed.</i>
49		Write a program to implement Linear Search	Hands- on Demo	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C, 2nd ed.</i>
50		Write a program to implement Binary Search	Hands- on Demo	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C, 2nd ed.</i>

51		Hashing –Hash tables	Interactive Lecture	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
52		Hash functions	Interactive Lecture	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
53		Discussion on various sorting methods, comparisons etc	Active Learning-Tutorial	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
54		Write a program to implement Insertion sort	Hands- on Demo	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
55		Write a program to implement Bubble sort	Hands- on Demo	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
56		Strategies to avoid collisions	Interactive Lecture	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
57		Strategies to resolve collisions	Interactive Lecture	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
58		Hashing using Quadratic Probing and Double Hashing	Active Learning-Tutorial	E. Horowitz and S. Sahni, <i>Fundamentals</i>

				<i>of Data Structures in C, 2nd ed.</i>
59		Write a program to implement hash table using Linear Probing	Hands- on Demo	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C, 2nd ed.</i>
60		Write a program to implement hash table using Quadratic Probing	Hands- on Demo	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C, 2nd ed.</i>
61		Binary Tree & its operations	Interactive Lecture	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C, 2nd ed.</i>
62		Implementation and its traversals	Interactive Lecture	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C, 2nd ed.</i>
63	5	Binary search Tree – Creation, Traversal, Searching and Deletion	Active Learning-Tutorial	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C, 2nd ed.</i>
64		Write a program to implement binary search tree creation	Hands- on Demo	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C, 2nd ed.</i>
65		Write a program to implement binary search tree traversals	Hands- on Demo	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C, 2nd ed.</i>

66		Binary Search Tree & its Operations	Interactive Lecture	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
67		Implementation and its traversals	Interactive Lecture	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
68		AVL Tree & its Operations	Active Learning-Tutorial	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
69		Write a program to implement AVL Tree and its Rotations	Hands- on Demo	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
70		Write a program to implement AVL Tree and its Rotations	Hands- on Demo	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
71		Implementation and its traversals	Interactive Lecture	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
72		B-Trees - Creation, Traversal, Searching and Deletion	Interactive Lecture	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C</i> , 2nd ed.
73		Discussion on B-Trees and its operations	Active Learning-Tutorial	E. Horowitz and S. Sahni, <i>Fundamentals</i>

				<i>of Data Structures in C, 2nd ed.</i>
74		Write a program to implement B-Trees – Creation	Hands- on Demo	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C, 2nd ed.</i>
75		Write a program to implement B-Trees – Searching and Deletion	Hands- on Demo	E. Horowitz and S. Sahni, <i>Fundamentals of Data Structures in C, 2nd ed.</i>

Assessment and Evaluation

Continuous Internal Evaluation (CIE) :70%	
Components of CIE	Weightage of each component
CIE-1 - Objective Type Graded Component	20 marks out of 70
CIE-2 - Theory	25 marks out of 70
CIE-3 - Lab Experiment - Viva - Completing Online Course,Evaluation and certificate Submission - Tutorial Submission & Evaluation	6 marks out of 70 4 marks out of 70 10 marks out of 70 5 marks out of 70

Semester End Exams (SEE): 30%	
Mode of Exam	Practical
Weightage of exam	30%

Course Outcomes- Programme Outcomes Matrix

Program Outcome Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1 Understand and explore the concepts of linear and non-linear data structures to develop solutions for diverse problems.	3	2	1	1	-	2	-	-	1	-	-	2
CO2 Apply the concepts of hashing, sorting and searching techniques to get an optimized solution.	3	2	1	1	-	2	-	-	1	-	-	2
CO3 Evaluate and apply appropriate data structure(s) for real-world problems.	3	2	1	1	-	2	-	-	1	-	-	2
CO4 Demonstrate Good Coding Practices involving life-long learning.	3	2	1	1	-	2	-	-	1	-	-	3

Rubrics for evaluation of CIE (separate for each CIE component)

Total Marks		20
CIE-1 Components		Rubrics for Assessment
CIE-1	Objective Type Graded Component (Conceptual and Programming)	Objective Type Questions will be conducted for 20 marks. Each right answer carries one mark each. No negative marking for wrong answers.

Total Marks		25
CIE-2 Components		Rubrics for Assessment
CIE-2	Theory	Test will be conducted for 25 marks and will be evaluated according to Scheme of Evaluation prepared by team of course faculty.

Total Marks	25
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CIE-3 Components		Rubrics for Assessment
CIE-3	Practical Component	Lab record+ Viva = 6+4 = 10marks (<i>Refer Table-3</i>)
	Online Course Module Completion	Course Module Completion - 10 Marks (<i>Refer Table-4</i>)
	Tutorial Submission and Evaluation	Execution + Results = 3+ 2 = 5marks (<i>Refer Table-5</i>)

Table-3

Lab Write-Up and Execution Rubrics (Max: 6 Marks)					
S.No.	Criteria	Measuring Methods	Excellent	Good	Poor
1	Understanding of the problem statement (2 Marks)	Lab Note Book	Student exhibits thorough understanding of the problem statement. (2 Marks)	Student has sufficient understanding of the problem statement. (1 Mark)	Student does not have a clear understanding of the problem statement. (0 Marks)
2	Execution and passing the test cases (2 Marks)	Lab Note Book	Student demonstrates the execution of the program with optimized code. Appropriate validations passing all the test cases. (2 Marks)	Student demonstrates the execution of the program without optimized code and passing only a few of the test cases. (1 Mark)	Student implemented but has not executed the program. None of the test cases passed. (0 Mark)
3	Results and Documentation (2 Marks)	Lab Note Book	Documentation with appropriate comments and output is covered in manual. (2 Marks)	Documentation with only few comments and few outputs are covered in manual. (1 Mark)	Documentation with no comments and no output cases is covered in manual. (0 Mark)

Viva-Voce Rubrics (Max: 4 Marks)

1	Conceptual Understanding (2 Marks)	Viva Voce	Explains thoroughly the solution along with related concepts. (2 Marks)	Adequately explains the solution and the related concepts (1Mark)	Unable to explain the solution (0 Mark)
2	Use of techniques (1 Mark)	Viva Voce	Insightful explanation of the appropriate data structure for the given problem to derive a solution (1 Mark)	Sufficient explanation of the appropriate data structure for the given problem to derive a solution (0.5 Mark)	Unable to explain the data structure for the given problem (0 Mark)
3	Communication of Concepts (1 Mark)	Viva Voce	Communicates the concept used in problem solving well. (1 Mark)	Sufficiently communicates the concepts used in problem solving. (0.5 Mark)	Unable to communicate the concepts used in problem. (0 Mark)

Table -4

Online Course Module Completion (Max: 10 Marks)							
S.No.	Criteria	Measuring Methods	Excellent	Very Good	Good	Fair	Poor
1	Course Completion Status (10 Marks)	Online Course Platform Module Completion	Student has successfully completed the entire course within the stipulated time without any pending components. (10-9 Marks)	Student has completed the entire course with minor delays in submission or reattempt. All mandatory requirements are eventually cleared. (8-7 Marks)	Student has completed most of the course content but with noticeable delays or multiple reattempts (6-5 Marks)	Student has partially completed the course with several modules or incomplete or with multiple extensions (4 Marks)	Student has registered for the course but has not completed the required modules (0-3 Marks)

Table-5

Tutorials submission and Evaluation Rubrics (Max: 5 Marks)					
S.No.	Criteria	Measuring Methods	Excellent	Good	Poor

1	Execution and passing the test cases (3 Marks)	Tutorials Submission in Hackerrank	Student demonstrates the execution of the program with optimized code. Appropriate validations passing all the test cases. (3 Marks)	Student demonstrates the execution of the program without optimized code and passing only a few of the test cases. (2 Mark)	Student implemented but has not executed the program. None of the test cases passed. (1 Mark)
2	Results and Documentation (2 Marks)	Tutorials Submission in Hackerrank	Documentation with appropriate comments and output is covered in manual. (2 Marks)	Documentation with only few comments and few outputs are covered in manual. (1 Mark)	Documentation with no comments and no output cases is covered in manual. (0 Mark)

Signature of the Course Faculty